

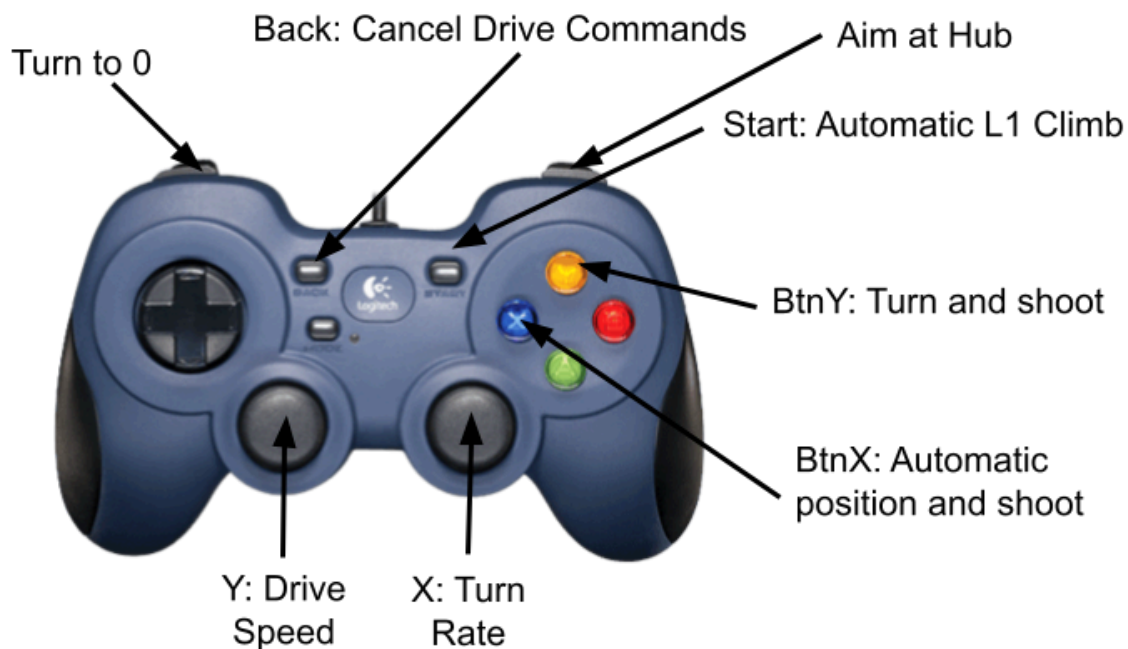
Recoil Robotics

2026 Robot User Manual

Version 1.1 (2/21/26)

Driver User Interface

The driver is responsible for controlling the robot's movement around the playing field. The driver uses one joystick attached to the driver station as controller 0.

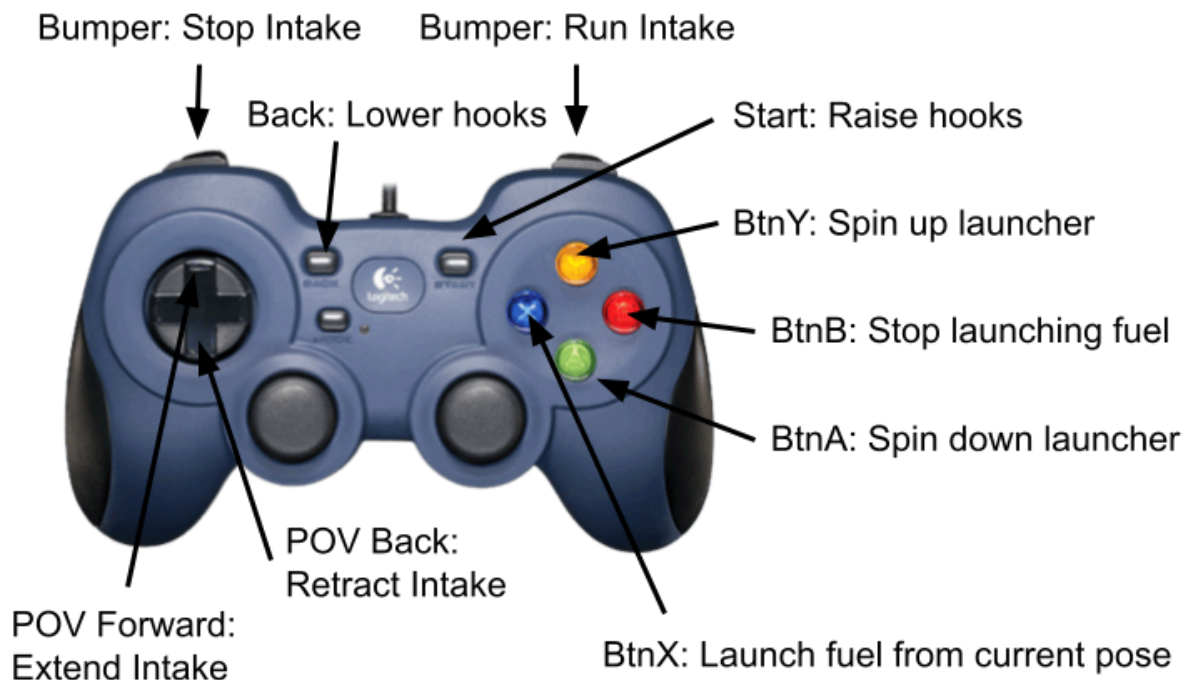


Control	Function	Control ID
Drive Speed	Forward/reverse speed	Gamepad Left Y
Turn Rate	Turn speed	Gamepad Right X
Turn and shoot	Spins up the flywheel to an appropriate rate relative to the robot's distance to the hub while rotating the robot to	Y

	face the hub, then shoots all of the fuel. Shooting ends when the Y button is released.	
Rotate to Hub	Turns the robot to point at the hub	Right Bumper
Cancel Drive Commands	Cancels all drive commands and returns to teleop (the default command)	Back / Joystick Input > 0.5

Operator User Interface

The operator is responsible for controlling non-drive-related subsystems on the robot and also has manual overrides for various automatic operations otherwise controlled by the driver.



SubSystem	Control	Function	Control ID
Scoring	Off	Spins down the flywheel and stops the lift motor	B

	Shoot manual	Shoots all of the fuel from the robot's current pose.	X
	Spin up flywheel	Starts the flywheel and track speed based on hub distance.	Y
	Stop flywheel	Spins down the launcher flywheel.	A
Intake	Extend	Extends the intake mechanism over the bumpers.	Pov Forward
	Retract	Retracts the intake mechanism over the bumpers and stops the intake motor if it's running.	Pov Backward
	Run Intake	Runs the intake motor.	Right bumper
	Stop Intake	Stops the intake motor.	Left bumper

Class to Command Mapping

The following table maps command operations described above to Java classes found in the “commands” subdirectory of the 2026 robot source code (see <https://github.com/frc9577/recoil/tree/main/2026-Rebuilt>).

Description	Class Name	Notes
Driver Commands		
Drive Speed & Turn Rate	N/A	Default DriveSubsystem command
Cancel drive commands	CancelDriveCommand	
Turn and shoot	Composite command see bindings	Composite command - turn to align on hub, calculate range, StartFlywheelCommand, StartLiftCommand when target speed reached and hub aligned.
Operator Commands		
Run intake	StartIntakeCommand	
Stop intake	StopIntakeCommand	
Extend intake	ExtendIntakeCommand	
Retract intake	RetractIntakeCommand	
Launch fuel from current pose	StartShootCommand	Starts lift, indexer, and bulk motors
Stop launching fuel	StopShootCommand	Stops the launcher lift, indexer, and bulk motors
TEST: Start launcher flywheel	StartFlywheelCommand	Temporary binding for test purposes only! Starts the flywheel at a fixed speed.

